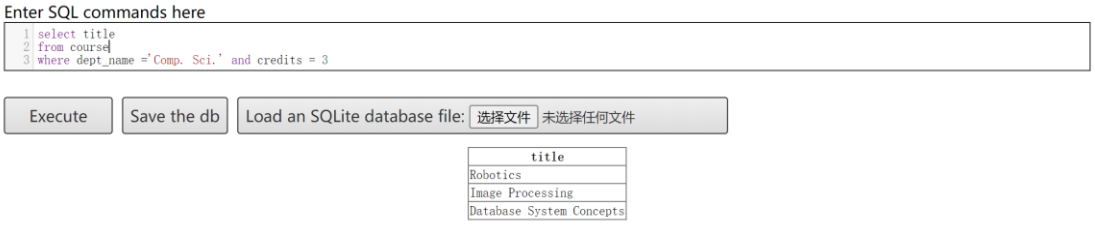
3.1

题目 a

select title
from course
where dept_name = 'Comp. Sci.' and credits = 3



题目 b

select distinct a.s_ID
from advisor as a, instructor as b
where a.i_ID = b.id and b. name = 'Einstein'



题目 c

select salary
from instructor
where salary >= all(select salary from instructor)

题目 d

select name
from instructor
where salary >= all(select salary from instructor)

题目 e

select title, count(*)
from takes, course
where takes.course_id = course.course_id and takes.semester = 'Fall' and takes.year = 2017

group by takes.course_id

Enter SQL commands here

```
select title, count(*)
from takes, course
where takes.course_id = course.course_id and takes.semester = 'Fall' and takes.year = 2017
group by takes.course_id

```

Execute | Save the db | Load an SQLite database file: 选择文件 未选择任何文件

title	count(*)
Intro. to Computer Science	6
Database System Concepts	2
Physical Principles	1

题目 f

```
select count(*)
from takes, course
where takes.course_id = course.course_id
and takes.semester = 'Fall'
and takes.year = 2017
group by takes.course_id
having count(*) >= all (select count(*) from takes, course
                        where takes.course_id = course.course_id
                        and takes.semester = 'Fall'
                        and takes.year = 2017
                        group by takes.course_id)
```

题目 g

```
select title
from takes, course
where takes.course_id = course.course_id
and takes.semester = 'Fall'
and takes.year = 2017
group by takes.course_id
having count(*) >= all (select count(*) from takes, course
                        where takes.course_id = course.course_id
                        and takes.semester = 'Fall'
                        and takes.year = 2017
                        group by takes.course_id)
```

3.3

题目 a

```
update instructor
set salary = 1.1*salary
where dept_name = 'Comp. Sci.'
```

题目 b

```
delete from course
where course_id not in(select course_id from section)
```

题目 c

```
insert into instructor(ID, name, dept_name,salary)
select ID, name, dept_name,10000
from student
where student.tot_cred>100
```

3.5

问题 a

```
select ID,(case
            when score<40 then 'F'
            when score>=40 and score <60 then 'C'
            when score>=60 and score <80 then 'B'
            when score >=80 then 'A') as class
from marks
```

问题 b

```
with classlevel(ID,class) as (select ID,(case
            when score<40 then 'F'
            when score>=40 and score <60 then 'C'
            when score>=60 and score <80 then 'B'
            when score >=80 then 'A') as class
from marks)
select count(*) as studentnum
from marks, classlevel
where marks.ID = classlevel.ID
group by classlevel.class
```

3.11

问题 a

```
select distinct a.ID, a.name
from student as a, takes as b, course as c
where a.ID = b.ID and b.course_id = c.course_id and c.dept_name = 'Comp. Sci.'
```

问题 b

```
select a.id, a.name
from student as a, takes as b, course as c
where a.id = b.id and b.course_id = c.course_id and a.id not in(select id from takes where
takes.year<2017)
```

问题 c

```
select dept_name,max(salary)
from instructor
group by dept_name
```

问题 d

```
with deptsalary(salary) as(select max(salary)
from instructor
group by dept_name)
select min(salary)
from deptsalary
```

3.13

```
person
(
   driver_id varchar(10)not null,
   name varchar(20)not null,
   address varchar(50),
   primary key(driver_number)
)
car
(
   license_plate varchar(15) not null,
   model varchar(10),
   year int,
   primary key(license_plate)
)
accident
(
   report_number varchar(30)not null,
   year int,
   location varchar(50)
   primary key(license_plate)
)
owns
(
   driver_id varchar(10)not null,
   license_plate varchar(15) not null,
   primary key(driver_id, license_plate),
   foreign key(driver_id)references person,
   foreign key(license_plate)references car
)
participated
```

```
(
  report_number varchar(30)not null,
  license_plate varchar(15)not null,
    driver_id varchar(10),
    damage_amount numeric(10,2)
    primary key(report_number, license_plate),
    foreign key(license_plate)references car,
    foreign key(driver_id) references person
)
```

3.14

问题 a

```
select count(*)
from person as a, owns as b, participated as c
where a.driver_id = b.driver_id and b.license_plate = c.license_plate and a.name = 'John Smith'
```

问题 b

```
update participateed
set damage_amount =3000
where report_number ='AR2197' and license_plate = 'AABB2000'
```

3.17

问题 a

```
update works
set salary = 1.1*salary
where company_name = 'First Bank Corporation' and ID in (select ID from employee)
```

问题 b

```
update works
set salary = 1.1*salary
where company_name = 'First Bank Corporation' and ID in (select ID from manages)
```

问题 c

```
delete from works
where company_name = 'Small Bank Corporation'
```

3.22

```
Where 1>=(select title from course)
```

3.24

```
select a.name, a.ID
from student as a, advisor as b, instructor as c
```

where a.id = b.s_id and b.i_id = c.id and c.dept_name = 'Philosophy'and a.dept_name = 'Accounting'

3.25

```
select dept_name
from department
where budget >(select budget from department where dept_name = 'Philosophy')
order by dept_name
```

3.31

```
select ID, name
from instructor
where instructor.id not in(
        select teaches.id
        from takes, teaches
        where teaches.course_id = takes.course_id
    and takes.grade = 'A'
)
```